Unit 2

Lesson 2

Innovators in Action.

Talk to Me!

Time Frame: [90 min]

Ready, Set, Learn!

You Need to Know

- Explain how a sensor sends a message to an app.
- Design or improve an app that speaks to the user.
- Write or organize simple instructions (code blocks) that control how the app behaves.
- Establish a Bluetooth connection between Arduino and the app to transmit sensor data.

Challenge questions of the lesson.

1. **Your ultrasonic sensor detects that something is very close—like a hand or a wall. If there's no screen, what is one way your app can tell the user what's happening using sound or speech?**
2. **If you're designing an app for someone who can't see the screen well, what are two features you could add so the app still gives them the right message at the right time?**
3. **Imagine your app should say a warning message only when something is detected by the ultrasonic sensor. What kinds of code blocks would you need to use to make that happen correctly?**

SEL Relation

1. **Empathy & Perspective-Taking (Self-Awareness & Social Awareness):**
   - Students design their app to speak messages aloud—often to help users who may have difficulty seeing or reading the screen. This requires them to step into the shoes of others, understand accessibility needs, and create supportive, inclusive solutions. They must ask themselves, "How can someone who can't see still know what's happening?"
2. **Responsible Decision-Making:**
   - Students make thoughtful choices about how the app should behave when **triggered** by sensor input. For example, they choose what messages are spoken, how often they should play, and what output is most helpful without being overwhelming. These decisions reflect real-world design thinking and ethical technology use.
3. **Relationship Skills & Collaboration:**
   - When testing sensor-app behaviour, students often work in pairs or teams. They practice clear communication, give constructive feedback, and coordinate roles during wiring, coding, or Bluetooth pairing. These teamwork moments strengthen their ability to collaborate respectfully and solve problems together.
4. **Growth Mindset & Perseverance:**
   - Setting up Bluetooth, writing condition-based code, and debugging app behaviour can be challenging. Students are encouraged to try, test, fail, and fix, building resilience and perseverance as they refine their designs. Teachers can reinforce the idea that problem-solving is a process, not a one-time task.

Engage

Teacher Role:

- Spark curiosity by reading aloud the dialogue between Adam, Laila, and Mrs. Sara to create a real-world connection.
- Facilitate a class discussion on Bluetooth and sensor-to-app communication by prompting students to think like developers.
- Scaffold by previewing key vocabulary such as Bluetooth, sensor, interactive, and testing.
- Use guiding questions like: “What would happen if the sensor didn’t work?” or “How would you know your app is really working?”
- Encourage empathy by linking the functionality of the app to how it could help someone with visual impairments.

Student Expectations:

- Actively listen to the story and identify what the characters are trying to achieve (connecting Arduino to the app).
- Answer and reflect on the Think questions:
  - o “What steps will you take to make the app interactive?”
  - o “Why is testing an app important before using it in real life?”
- Demonstrate understanding by discussing how Bluetooth enables sensor data to be transferred to the app.
- Engage in paired discussions to build on each other's ideas and connect story to their own project development.

Parts of the books included.


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The next day in the class, Adam and Laila worked on connecting the Bluetooth module with the help of Ms. Sara.

Ms. sara connected the module to the Arduino. "Now, let’s pair it with the phone."

Laila opened the app. "I’ll test the connection. Adam, can you trigger the sensor."

Adam pressed a button. "Is the app getting the data?"

Laila smiled. "Yes! Now, let’s make the app speak the distance."

Adam typed quickly. "I’m adding the code to send the data to the app."

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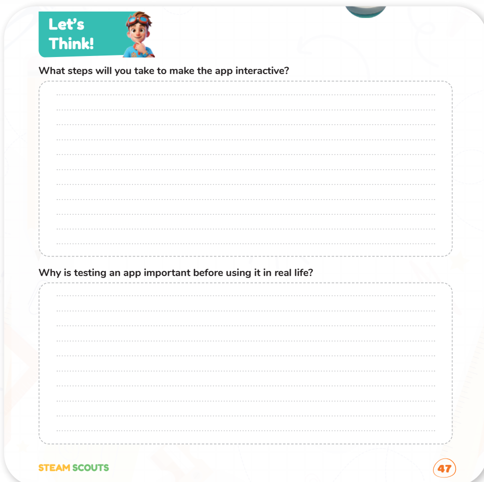
Let’s Think!

What steps will you take to make the app interactive?

Why is testing an app important before using it in real life?

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Story

- Welcome the students and let them know today's challenge is about helping an app "speak" sensor data from an Arduino device.
- Introduce the characters (Adam, Laila, and Mrs. Sara) and display the story Scene 1 on the board or read it aloud dramatically.
- Use expressive voice, pausing at key moments to ask, "What do you think is happening here?" or "Why is Laila testing the app connection?"
- Emphasize how the students in the story are working as a team to connect a sensor to an app using Bluetooth—and that your students will be doing the same!

Possible Strategies for Implementation:

1. **Visible Thinking Routine – "See, Think, Wonder"**
   - See: Ask students to describe what they see happening in the story.
     - o "What tools are the students using?"
   - Think: Encourage them to analyze.
     - o "Why did Laila want to test the connection before moving on?"
   - Wonder: Promote curiosity.
     - o "What do you wonder about making an app speak using code?"
2. **Kagan Strategy – "Think–Pair–Share"**
   - After reading the story, pose the question:
     - o "If you were Laila or Adam, what step would you take next to finish the project?"
   - Let students think quietly, then share ideas with a partner.
   - Invite pairs to share out with the class and chart their ideas.
3. **Anchor the Learning with a Real-Life Connection.**
   - Prompt students:
     - o "Have you ever used an app that talks? How do you think it works?"
   - Lead into:
     - o "Today, you'll build your own app that speaks when a sensor detects something!"

Let's Think!

Objective:

- Students will reflect on how to design interactive, functional mobile applications by identifying key steps and understanding the importance of testing for real-world usability.

Answers:

1. **What steps will you take to make the app interactive?**
   - Add buttons or sensors to the app, use blocks that detect sensor input, add text-to-speech so the app talks, and test if each part responds when triggered.
2. **Why is testing an app important before using it in real life?**
   - Testing helps find problems, ensures the app works as expected, and makes it safer and easier for people to use, especially if it supports someone with a disability.

Possible Strategies for Implementation:

1. **Think-Pair-Share:**
   - Ask students to first reflect quietly and write down their answers. Then, pair with a classmate to discuss their responses. Finally, invite a few pairs to share with the class. This supports collaboration and deeper reasoning.
2. **Visible Thinking Routine – "Claim, Support, Question":**
   - Guide students to state a claim (e.g., "Testing is important"), support it with evidence ("because it might not work correctly the first time"), and then ask a thoughtful question (e.g., "What if the user doesn't hear the sound from the app?"). This builds reasoning and ownership of learning.
3. **Anchor to Real-World Use:**
   - Briefly show or describe a real-life app that uses sensors and explain how it needed testing before being released (e.g., a step counter or talking calculator). Ask: "What could go wrong if it wasn't tested properly?"

Explorer's Toolkit

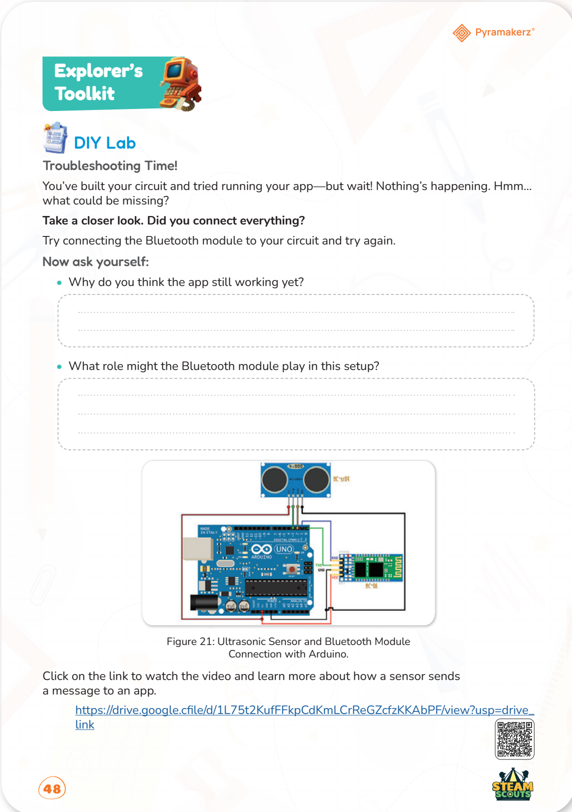
Teacher Role:

- Guide students in building the circuit correctly and safely.
- Prompt students to reflect on their troubleshooting steps.
- Help connect the physical setup to the app functionality.
- Encourage curiosity and support when things don't work the first time.
- Facilitate discussion after the video to reinforce understanding.

Expected from Students:

- Follow the diagram and assemble the circuit with attention to detail.
- Use critical thinking to troubleshoot why the app isn't receiving data.
- Answer reflective questions thoughtfully.
- Engage with the video to connect practical experience with conceptual understanding.

Parts of the books included.



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Explorer's Toolkit

DIY Lab

Troubleshooting Time!

You've built your circuit and tried running your app—but wait! Nothing's happening. Hmm... what could be missing?

Take a closer look. Did you connect everything?

Try connecting the Bluetooth module to your circuit and try again.

Now ask yourself:

- Why do you think the app still working yet?
- What role might the Bluetooth module play in this setup?

Figure 21: Ultrasonic Sensor and Bluetooth Module Connection with Arduino.

Click on the link to watch the video and learn more about how a sensor sends a message to an app.

https://drive.google.cfile/d/1L75t2KufFFkpCdKmLCrReGZcfzKKAbPF/view?usp=drive_link

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DIY Lab

Objective:

Students will investigate how the Bluetooth module allows data to travel from the sensor to the app, and troubleshoot their physical computing setup to ensure proper communication between components.

Answers:

1. **Why do you think the app still isn't working yet?**
   - Because the Arduino hasn't been programmed to send any data to the Bluetooth module. Without uploading the correct code, the sensor won't collect data or send it to the app.
2. **What role might the Bluetooth module play in this setup?**
   - The Bluetooth module acts like a messenger. It takes the data from the Arduino (which receives it from the ultrasonic sensor) and sends it wirelessly to the mobile app so it can be read or spoken aloud.

Materials Needed:

- Pre-assembled Arduino circuit.
- Laptop with Arduino IDE installed.
- MIT App Inventor mobile app (design created in the previous lesson).
- Android device with Bluetooth capability.

Activity Instructions:

1. Students begin by examining the provided circuit diagram with a focus on the Bluetooth module's role.
2. The teacher or students connect the Bluetooth module to the pre-built circuit.
3. Students observe what happens in the app interface and discuss:

- Is the app receiving data now?
- What does the Bluetooth module do in this system?

4. Students then respond to two reflection questions in their workbook:

- Why do you think the app wasn't working before?
- What role does the Bluetooth module play?

Possible Strategies for Implementation:

- See–Think–Wonder (Visible Thinking): After observing the setup, students write what they see, think is happening, and wonder about the Bluetooth role.
- Think–Pair–Share: Students reflect individually on the reflection questions, discuss their thoughts with a partner, and share key takeaways with the class.
- Teacher Prompted Demo: Conduct the Bluetooth connection as a live demo while asking students to predict what will happen at each step.
- Mini-Discussion Circles: Small groups rotate through the station, connect the module, observe results, and discuss before answering the questions.

Explain

Teacher Role:

- Introduce the topic by connecting it to students' earlier hands-on experience.
- Ask guiding questions before the video to set a clear purpose for watching.
- Pause during key moments of the video to check understanding.
- Facilitate a short discussion after the video to reinforce connections between sensors, Bluetooth, and app feedback.
- Clarify technical terms and ensure students relate them to the class setup.

Expected From Students:

- Watch the video attentively, focusing on how the sensor sends information to the app.
- Identify the role of the Bluetooth module in the communication process.
- Ask questions or share what confused them.
- Use the video content to answer reflection questions or support their design choices in later activities.

Parts of the books included.

Click on the link to watch the video and learn more about how a sensor sends a message to an app.

https://drive.google.cfile/d/1L75t2KufFFkpCdKmLCrReGZcfzKKAbPF/view?usp=drive_link

Screen Lessons

Objectives:

Understand how sensor data is transmitted from Arduino to a mobile application using a Bluetooth module.

Video Link: https://drive.google.com/file/d/1L75t2KufFFkpCdKmLCrReGZcfzKKAbPF/view?usp=drive_link

Possible Strategies for Implementation:

1. **Pre-Watching:**
   - Activate prior knowledge by asking:
     - o "What did we connect earlier today?"
     - o "What do you think the Bluetooth module does when it's added?"
   - Introduce vocabulary like Bluetooth, sensor data, and communication.
2. **During Watching:**
   - Use a Kagan "Pairs Compare" pause point:
     - o Let students turn to a partner halfway through and discuss:
   1. "What part of the system is collecting the data?"
   2. "What part is sending the data?"
3. **Post-Watching:**
   - Lead a Visible Thinking Routine: Connect – Extend – Challenge.
     - o Connect: "What did you already know about this process?"
     - o Extend: "What new ideas or steps did the video add?"
     - o Challenge: "What's still confusing or tricky about how the sensor talks to the app?"

Elaborate

Teacher Role:

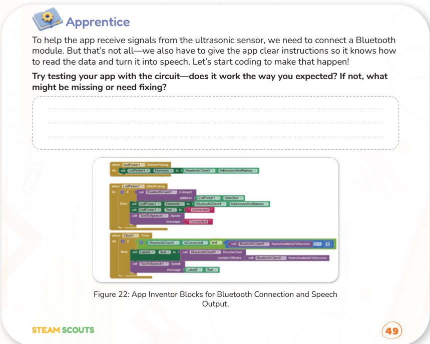
- Guide students through the purpose of the video by previewing key vocabulary (e.g., "Bluetooth module," "text-to-speech").

- Set a clear learning intention: "We're learning how sensors and apps communicate."
- Encourage active observation by prompting students to jot down or sketch what happens in the app when the sensor is triggered.
- Facilitate discussion after watching to clarify misunderstandings and connect observations to prior lessons.

Expected From Students:

- Watch the video carefully and identify the communication flow: sensor → Arduino → Bluetooth → app.
- Note what visual or verbal signals the app receives from the circuit.
- Participate in a follow-up discussion or Think-Pair-Share to explain how data moves from the circuit to the app.

Parts of the books included.


Apprentice

To help the app receive signals from the ultrasonic sensor, we need to connect a Bluetooth module. But that's not all—we also have to give the app clear instructions so it knows how to read the data and turn it into speech. Let's start coding to make that happen!

Try testing your app with the circuit—does it work the way you expected? If not, what might be missing or need fixing?

Figure 22: App Inventor Blocks for Bluetooth Connection and Speech Output.

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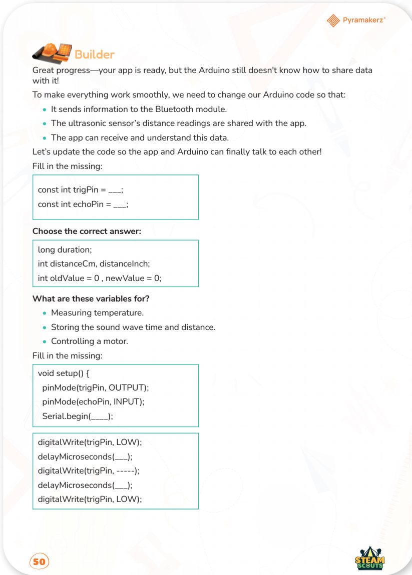
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Builder

Great progress—your app is ready, but the Arduino still doesn't know how to share data with it!

To make everything work smoothly, we need to change our Arduino code so that:

- It sends information to the Bluetooth module.
- The ultrasonic sensor's distance readings are shared with the app.
- The app can receive and understand this data.

Let's update the code so the app and Arduino can finally talk to each other!

Fill in the missing:

```
const int trigPin = ___;
const int echoPin = ___;
```

Choose the correct answer:

```
long duration;
int distanceCm, distanceInch;
int oldValue = 0 , newValue = 0;
```

What are these variables for?

- Measuring temperature.
- Storing the sound wave time and distance.
- Controlling a motor.

Fill in the missing:

```
void setup() {
 pinMode(trigPin, OUTPUT);
 pinMode(echoPin, INPUT);
 Serial.begin(____);
```

```
digitalWrite(trigPin, LOW);
delayMicroseconds(___);
digitalWrite(trigPin, -----);
delayMicroseconds(___);
digitalWrite(trigPin, LOW);
```

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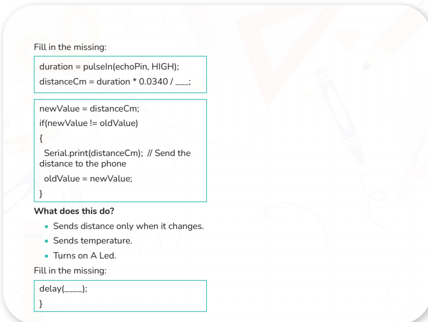

Fill in the missing:

```
duration = pulseIn(echoPin, HIGH);
distanceCm = duration * 0.0340 / ___;
```

```
newValue = distanceCm;
if(newValue != oldValue)
{
 Serial.print(distanceCm);  // Send the distance to the phone
 oldValue = newValue;
}
```

What does this do?

- Sends distance only when it changes.
- Sends temperature.
- Turns on A Led.

Fill in the missing:

```
delay(____);
}
```

Apprentice

Objective:

Build and test the App Inventor code to receive data from the ultrasonic sensor through Bluetooth and convert it into spoken feedback using Text-to-Speech.

Answers:

1. **Does it work the way you expected?**
   - “Yes! After adding the code blocks in App Inventor and connecting to the Bluetooth, the app read the distance data aloud when something was placed in front of the sensor.”
2. **If not, what might be missing or need fixing?**
   - “If it didn’t work, I would check if:
     - The Bluetooth module is connected in the app (ListPicker not selected)
     - The sensor isn't sending readable text

- The Text-to-Speech block isn't linked correctly in the Timer block"

Activity Instructions:

1. Open MIT App Inventor and create a new project.
2. Add the required components: ListPicker, Clock, BluetoothClient, Label, and TextToSpeech.
3. Drag and connect the code blocks (as shown in the image) to:
   - Connect to the Bluetooth device.
   - Receive serial data.
   - Speak the text received.

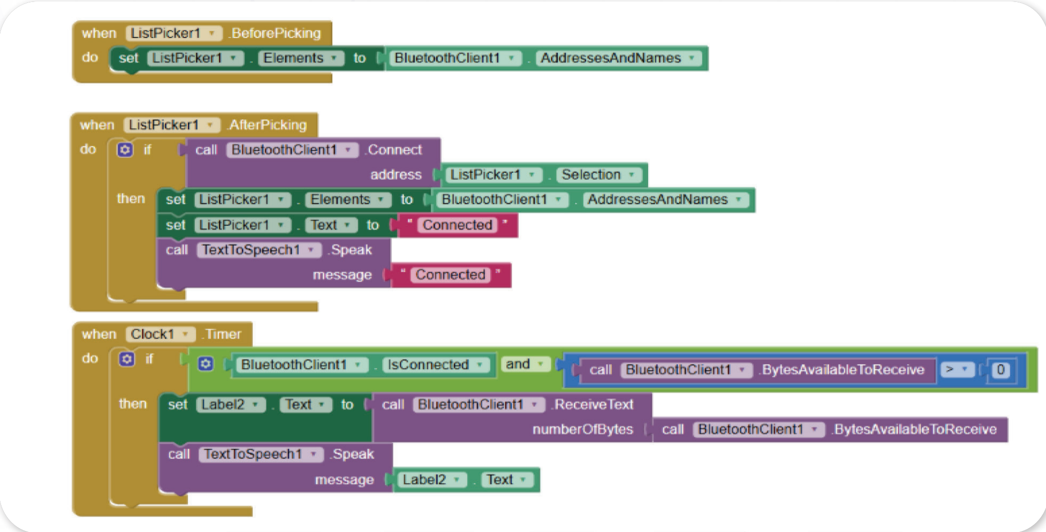


Figure 22: App Inventor Blocks for Bluetooth Connection and Speech Output.

4. Pair the mobile device with the HC05- Bluetooth module.
5. Test the app by placing an object near the ultrasonic sensor.
6. Observe if the app speaks the distance.
7. Answer the questions about what worked and what didn't.

Possible Strategies for Implementation:

- **Guided Build First**: Walk through the block setup together as a class using a shared screen or printed block sheet.
- **Partner Debugging:** Students pair up and check each other's blocks before testing.
- **Step-by-Step Verification:** Pause after each section of the code (ListPicker, Timer, TTS) and predict what it should do.
- **Reflection Corner:** Create a "What didn't work" board for students to post anonymous common mistakes or bugs they encountered during testing—foster a safe, collaborative debugging mindset.

Builder

Objective:

Objective: Students will write and complete an Arduino code that allows the ultrasonic sensor to measure distance and transmit it via the Bluetooth module to the mobile app, which will then speak the detected distance.

Answers:

1. Fill in the Missing:

```
const int trigPin = 9;
const int echoPin = 10;
```

2. Choose the correct answer:

```
long duration;
int distanceCm, distanceInch;
int oldValue = 0, newValue = 0;
```

- Answer:

B. Storing the sound wave time and distance.

3. Fill in the Missing (Setup):

```
void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
}
```

4. Fill in the Missing (Trigger Pulse):

```
digitalWrite(trigPin, LOW);
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);
```

5. Fill in the Missing (Distance Calculation):

```
cpp
CopyEdit
duration = pulseIn(echoPin, HIGH);
distanceCm = duration * 0.0340 / 2;
```

6. What does this do?

```
newValue = distanceCm;
if(newValue != oldValue) {
  Serial.print(distanceCm); // Send the distance to the phone
  oldValue = newValue;
}
```

- Answer:

A. Sends distance only when it changes.

7. Final Fill in the Missing:

```
delay(100);
```

Activity Instructions:

1. Warm-Up Discussion: Briefly review what the ultrasonic sensor measures and how Bluetooth modules send data.
2. Start Coding: Begin by writing the Arduino code using the prompts:
   - Define trigger and echo pins.
   - Set up the pin modes and serial communication.
   - Measure distance using pulseIn() and convert it to centimeters.
   - Compare old vs. new values and only send updated values via Serial.print.
   - Add delay to control frequency of updates.
3. Code Completion Tasks:
   - Fill in all blanks in the provided code scaffold.
   - Choose correct answers for embedded quiz questions.
4. Upload and Test: Upload the final code to Arduino and open the mobile app to test live communication.
5. Debug and Reflect: If it doesn't work, check connections and revisit missing steps. Update your code if necessary.

Possible Strategies for Implementation:

- Think-Pair-Share: Students fill in code blanks individually, then pair up to compare logic and debug.
- Visible Thinking Routine – "What Makes You Say That?": Ask students to justify each line they added or corrected in the code.
- Error Hunt: Provide an Arduino code with 3 hidden bugs. Students must find and fix them before using the correct version.
- Model First: Display a sample working code on the projector but hide some values. Ask guiding questions to help students understand what each part does.
- Pair Programming: Students take turns as "driver" and "navigator" as they write and test code together.

Evaluate

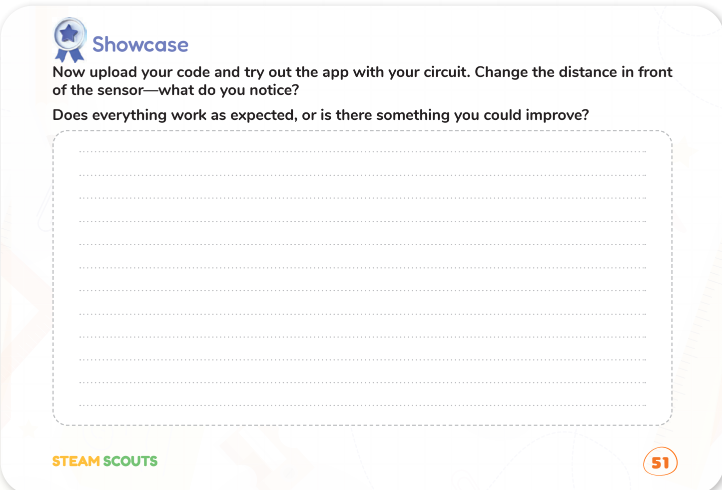
Teacher Role:

- Observe and support students as they run the complete system (sensor + Arduino + Bluetooth + App).
- Prompt them to notice how distance changes affect the app's spoken messages.
- Encourage students to identify potential issues or limitations and propose improvements.
- Guide reflective discussion about system behavior and debugging.

Expected From Students

- Upload their Arduino code and run the circuit with the connected app.
- Test the system by moving objects closer/farther from the ultrasonic sensor.
- Listen and observe the app's response (via Text-to-Speech).
- Reflect on whether the system works accurately and suggest any improvements.

Parts of the books included.


Showcase

Now upload your code and try out the app with your circuit. Change the distance in front of the sensor—what do you notice?

Does everything work as expected, or is there something you could improve?

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Showcase

Objective:

Students will test their full prototype (Arduino + Bluetooth + App) and evaluate its performance by observing how well the distance readings are transmitted and converted to speech.

Answer:

- "When I moved my hand closer, the app said a lower number. When I moved it farther, it said a higher number. Everything worked, but sometimes it repeated the number too often. I could add a pause or smoother reading next time."

Project Resources link: https://docs.google.com/document/d/1xCnRHZX0X894v7aR_BwtntuEa0xWLMSa/edit

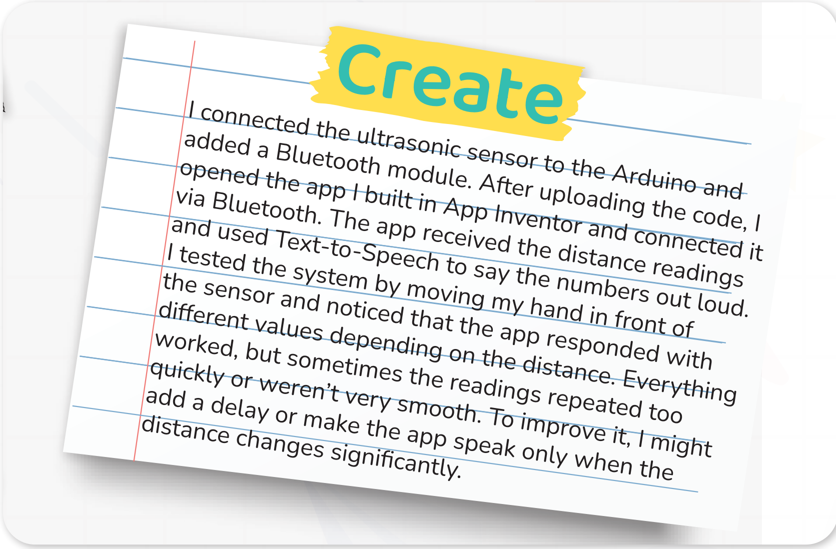
Activity Instructions

1. Upload your final Arduino code to your board.
2. Open your connected mobile app and select the Bluetooth device.
3. Move an object back and forth in front of the sensor.
4. Observe how the readings are spoken.
5. Write down what worked well and what could be improved.

Showcase Rubric:

Criteria	Excellent (4 Point)	Proficient (3 Points)	Developing (2 Points)	Needs Work (1 Point)
Functionality of Prototype.	Fully functional	Mostly working	Partially working	Not working
Accuracy of Response.	Accurate & real-time.	Generally accurate	Delayed response	Inaccurate values
Reflection and Improvement.	Detailed improvement ideas.	Clear insight	Vague observation	No reflection

It's time to move to the Create part and add this.



Now I Can

Objective:

Students will reflect on how they applied their understanding of smart technology, sensor communication, and app development by building a Bluetooth-connected talking app that receives data from Arduino.

Activity Instructions:

1. Introduce the Purpose.

- Say: "Now that we've completed our smart talking app and connected it with the Arduino, let's take a moment to think about all the skills we used and what we understand now."

2. Explain the Reflection Goal.

- Explain that this activity helps students recognize the steps they took to connect devices, code logic into their app, and use real-time sensor data in a meaningful way.

3. Read Each Statement Aloud.

- Read each Now I Can statement slowly and clearly:.
- Pause after each one and give students a moment to think.

4. Student Response Options.

- Have students check the box next to each statement they feel confident about.

5. Optional Sharing

- Let students share one "Now I Can" statement with a partner or in a small group.
- You may ask:
    - o "Which part of the talking app project helped you learn the most?"
    - o "What was tricky about connecting the Bluetooth, and how did you solve it?"
    - o "What was exciting about making the sensor speak using the app?"

6. Wrap-Up Reflection

- Ask students to look at their checklist and reflect:
  - o "Which coding or wiring skill would you like to practice more?"
  - o "How do you feel about building a smart system that speaks sensor data aloud?"

Relation to Standards:

1. ISTE Standards for Students (International Society for Technology in Education).

- **Empowered Learner:**
  - o Students take an active role in choosing and customizing their app interface and testing real-time sensor interaction.
- **Innovative Designer:**
  - o They use the Engineering Design Process to solve the real-world challenge of creating a talking app that receives and interprets sensor data.
- **Computational Thinker:**
  - o Students break down the system (sensor → Arduino → Bluetooth → app → speech) and create a functioning algorithm using logical sequences.

2. NGSS (Next Generation Science Standards).

- **MS-PS4-3** (Apply scientific ideas to design, test, and modify a device that converts signals to sound or light):
  - o The project converts sensor readings (distance) into audible speech using a TTS system, directly applying this performance expectation.
- **MS-ETS1-2** (Evaluate competing design solutions using a systematic process):
  - o Students troubleshoot their circuit and app communication, deciding which changes result in better accuracy or speed.

3. CSTA K–12 Computer Science Standards.

- **3A-AP-22** (Design and develop computational artifacts working in team settings):
  - o Students collaboratively build both the hardware (Arduino + sensor) and software (App Inventor blocks).
- **3A-AP-17** (Decompose problems and subproblems into parts to facilitate the design, implementation, and review):
  - o The entire system is broken into subcomponents: sensor data, transmission via Bluetooth, app logic, and TTS output.

4. Cambridge ICT Starters / Australian Digital Technologies Curriculum

- Creating Solutions by Creating Algorithms and Implementing Digital Systems:
  - o Students implement a real-world digital solution using sensors, microcontrollers, and app design, showcasing integration between hardware and software.
- Recognizing and Using Data:
  - o They interpret and use numeric distance data from the sensor, convert it into speech, and troubleshoot data reception via Bluetooth.

5. UNESCO ICT Competency Framework for Teachers

- Technology Literacy: Basic hardware/software operations and concepts:
  - Students demonstrate understanding of input/output systems, sensors, and Bluetooth communication in a hands-on context.
- Knowledge Deepening: Using ICT to solve complex problems:
  - The project encourages students to think critically about system communication, synchronization, and usability for accessibility purposes.